Endmember-Guided Spectral–Spatial Feature Diffusion for Hyperspectral Anomaly Detection

Caihuan Huang, Li Liu, Chao Xiao, Qiang Ling, Guosheng Li, and Kun Li

***Abstract*—Hyperspectral anomaly detection (HAD) aims at identifying the pixels whose spectral features notably deviate from their surroundings. However, when the spectral curves of targets are mixed with those of background in a pixel, existing HAD methods suffer from significant performance degradation due to the subtle spectral differences between targets and background. In this article, we propose an endmember-guided feature diffusion network to address this problem. Specifically, a diffusion network is first proposed to regularize the spectral–spatial features of hyperspectral images (HSIs) extracted by an autoencoder. Afterward, an unmixing module is designed to obtain the endmember features to represent the prototypes of HSIs. The proposed diffusion model subsequently aggregates the endmember features to update the HSIs features according to the spectral linear mixing model. Finally, the feature learning and diffusion are jointly optimized to ensure that the proposed method can deal with the negative effect of mixed pixels on HAD. Experimental results demonstrate that the proposed method has superior detection performance to several state-of-the-art HAD methods. In particular, when the target abundance is 0.01 in mixed pixels on the synthetic dataset, our method can better identify the anomaly targets.**

***Index Terms*—Anomaly detection, autoencoder, diffusion, hyperspectral image (HSI), spectral unmixing, subpixel target.**

I. Introduction

HYPERSPECTRAL images (HSIs) have nanoscale spectral resolution. Each pixel in an HSI can be represented by a continuous spectral curve, enabling the discrimination of subtle differences between various materials. Therefore, HSIs have been widely applied in various domains, including remote sensing, military reconnaissance, and environmental monitoring. Hyperspectral anomaly detection (HAD) [1] aims to identify pixels whose spectral curves significantly deviate from their local surroundings. However, this task remains challenging due to the lack of prior spectral knowledge about the targets.

Existing HAD methods identify anomalies based on differences in spectral features. Traditional methods [2], [3], [4], [5] model the background as statistical distributions. Pixels that deviate from these distributions are considered anomalies. Moreover, since anomalies usually occur with low probability, they exhibit a sparse characteristic, whereas the background generally has a low-rank structure. Several approaches [6], [7], [8] have utilized these priors to enhance anomaly detection performance. Recently, deep learning-based methods [9], [10], [11], [12], [13], [14], [15] have emerged, which detect anomalies by analyzing reconstruction errors between the input and output of autoencoders or related architectures. However, mixed pixels are ubiquitous in HSIs due to limited spatial resolution of hyperspectral sensors. When targets appear as the subpixel form, the spectral differences between the targets and the background diminish, making anomalies more difficult to be distinguished from the background. Consequently, the detection performance of most existing methods significantly degrades for the presence of mixed pixels. As shown in Fig. 1, the AUC values achieved by existing methods notably decrease when the dataset contains subpixel targets.

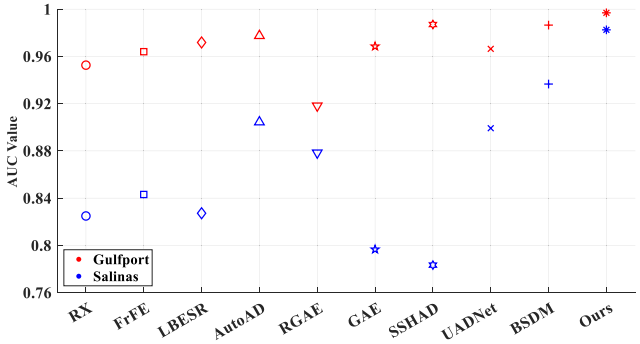

Fig. 1. Comparison of the negative effect of mixed pixel on AUC values achieved by different methods. The targets on the Gulfport dataset appear as full-pixel form (colored in red), while the targets on the Salinas dataset appear as subpixel form (colored in blue).

Hyperspectral unmixing [16] is used to analyze the spectral components of HSIs to address this issue. Nevertheless, when target abundances are extremely low within mixed pixels, these methods [13], [17], [18] still suffer from significant performance degradation. This is mainly because these methods only rely on spectral features of HSIs while ignoring the spatial information that represents the continuity of the background. Moreover, the feature extraction capabilities of these models are limited, making it difficult to distinguish anomalies from the background. Furthermore, since hyperspectral unmixing and feature learning are usually performed separately in most existing methods, the cumulative errors will produce negative impact on detection performance.

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Caihuan Huang, Guosheng Li, and Kun Li are with the College of Big Data and Information Engineering, Guizhou University, Guiyang 550025, China (e-mail: gs.chhuang24@gzu.edu.cn; gs.gsli24@gzu.edu.cn; lik@gzu.edu.cn).

Li Liu, Chao Xiao, and Qiang Ling are with the College of Electronic Science and Technology, National University of Defense Technology, Changsha 410073, China (e-mail: liuli_nudt@nudt.edu.cn; xiaochao12@nudt.edu.cn).

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To tackle the aforementioned problems, we propose an endmember-guided spatial–spectral feature diffusion (EG-SSFD) network for HAD. Unlike original denoising diffusion probabilistic models (DDPMs) [19], [20], [21], we train the model in the feature space instead of the pixel space. Specifically, we first perform principal component analysis (PCA) on HSIs to reduce spectral redundancy while preserving critical signal energy. Then, a lightweight convolutional autoencoder (ConvAE) is designed to extract spectral–spatial features from the HSI patches, which are obtained by applying sliding window to the compressed HSIs. A Transformer-based diffusion model is proposed to regularize the extracted features due to its powerful generative capability. Moreover, we design a hyperspectral unmixing module to learn endmember priors, and integrate them into the denoising network to update the spectral–spatial features via cross-attention mechanism. Our method is trained in an end-to-end manner by coupling feature extraction and feature diffusion, which reduces the cumulative errors between them and ensures that the learned representations adapt to the anomaly detection task. Therefore, our method remarkably improves both robustness and accuracy.

The main contributions of this article are summarized as follows.

1) We propose a spatial–spectral feature diffusion framework for HAD. Specifically, both spatial and spectral features are learned to integrate into the diffusion process by a novel noise estimation network that is applicable to the HAD task.
2) We introduce endmember priors to guide the spatial–spectral feature diffusion, thereby enhancing feature representation. This enables our method to effectively handle subpixel target detection, which is particularly challenging in real-world scenarios.
3) We jointly optimize feature extraction and diffusion in an end-to-end manner to improve detection accuracy. Unlike traditional diffusion model-based training paradigm, we combine the reconstruction loss with diffusion loss to train the network, and thus reduce the cumulative errors between the two processes.

We organize the remainder of this article as follows. In Section II, we briefly review the related works. In Section III, we introduce our method in detail. We present and analyze the experimental results in Section IV, and finally conclude this article in Section V.

II. Related Work

A. Hyperspectral Anomaly Detection

Since HAD is an unsupervised detection task, background modeling plays an important role in HAD [22], [23], [24]. Statistical-based methods such as Reed–Xiaoli (RX) [2], and its variants [3], [25] model the background as a specific distribution (e.g., multivariate Gaussian distribution). The pixels that deviate from the distribution are detected as anomalies. In [26], the spatial information is used to obtain various background feature distributions. These methods have high computational efficiency, but they notably depend on the accuracy of the background distribution assumption. However, the complex background may not follow any specific statistical assumption. Therefore, several methods perform HAD in other complex spaces such as kernel space [25], [27], frequency space [4], and topological space [5] by introducing domain transform. Background follows specific distribution on these space. Moreover, spectral angular distance, Euclidean distance, and local outlier factor can measure the degree of spectral differences between background and anomaly. The pixels with large distance from the background are identified as anomalies [28], [29]. In [30], graphical data description is used to adaptively analyze anomalies without distribution assumption.

Representation-based methods [7], [8], [31], [32], [33], [34] represent background as a linear combination of background atoms. Since anomalies cannot be linearly represented by the background dictionary, they have larger representation residuals than background. According to different constraints on the representation coefficients, these methods are divided into sparse representation-based methods, collaborative representation-based methods, and low-rank representation-based methods. Specifically, sparse representation-based methods [7], [32], [35], [36] suppose that anomalies are sparse due to their low occurrence, and calculate the representation residuals by sparse regression. Collaborative representation-based methods [8], [31], [37] assume that the local neighborhood pixels should participate in representing the central pixels. These methods use l_2-norm on the coefficients and obtain the closed-form solution. However, the aforementioned methods only use the local features of HSIs, while the global features are not exploited.

Low-rank representation-based methods [6], [18], [38], [39], [40] use low-rank characteristic to model background, and impose nuclear norm on representation coefficients. Low-rank constraint reflects the global property of HSIs. Therefore, these methods simultaneously employ local and global features of HSIs to detect anomaly targets. Existing methods [6], [18], [41] focus on constructing background dictionary to represent the background. The more pure dictionary can obtain better background representation. Moreover, structure features of HSIs are extracted to improve the representation capability of the models [42]. However, since these methods preprocess each HSI as a matrix for anomaly detection, it will lose the HSI cube structure. Tensor decomposition-based methods [43], [44], [45], [46] are proposed to preserve the 3-D cube information of HSIs. Although these methods achieve promising results, they depend on the artificial features that will limit their applications.

Deep learning-based methods [47], [48], [49], [50], [51], [52], [53], [54], [55], [56] usually use autoencoder and its variants to reconstruct HSIs. Since anomaly samples are much more than background samples, anomaly samples contribute less than background samples during network training. Test samples with larger reconstruction errors are classified as anomalies. However, the autoencoder cannot model the distribution of latent representation. Therefore, variation regularization [57], adversarial training [9], [10], and normalization flow [58] are introduced to regularize the latent representation to follow specific statistical distributions. Moreover, since autoencoder and its variants have powerful generalization capabilities, anomalies can also be well constructed during training [59]. There are three commonly used

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approaches to suppress anomaly generalization. Background purification selects pseudo background samples [12], [13] to train networks. Background masks [11], [60] are constructed to highlight background positions while suppress anomaly positions. Background memory [48], [61] is used to store the prototypes of background classes. The methods mentioned above mainly use spectral features. Since background is spatially homogeneous, feature fusion approaches [12], [62] are proposed to fully exploit the spatial features of HSIs. In addition, the frequency domain information [4], [63], [64] is integrated to enhance the feature representation. Due to the limited training samples, the detection performance of aforementioned methods may notably degrade when being tested on other scenes. Therefore, transfer learning [65], [66], [67], few-shot learning [68], and hyperspectral foundation model [69] are introduced to improve the generalization capability across domains.

Besides, considering the interpretability of deep learning-based methods, several methods [70], [71], [72], [73] combine the model-driven methods with data-driven methods to leverage their advantages.

B. HAD With Unmixing

Existing methods introduce hyperspectral unmixing [74] to relieve the negative effect of mixed pixels on HAD. The abundance map-based methods [17], [75], [76] perform HAD in the abundance space, where the feature contrasts between anomalies and background are more distinct. The endmember-based methods [13], [18] assume that background can be linearly represented as a combination of background endmembers, while anomalies cannot. Reconstruction is used to identify anomalies. However, due to the limited feature representation, these methods cannot achieve optimal detection results on subpixel targets especially when the abundances are extremely low.

C. HAD With DDPM

Since background suppression diffusion model (BSDM) [14] introduced DDPM for HAD, diffusion-based methods [49], [77] have attracted increasing attention in HAD due to its powerful generative capability. These methods use DDPM to constrain the background to follow the Gaussian distribution, and enhance the representation capability of models. However, these methods are trained in the pixel space, which have lower running efficiency. Moreover, these methods cannot exploit the spatial feature of HSIs so that they achieve unsatisfactory results when HSI background is complex. Finally, these methods are not designed for subpixel target detection and may fail to deal with mixed pixels.

III. Method

A. Overview

It is challenging to identify subpixel targets with extremely low abundances in mixed pixels. In this article, we propose an endmember-guided spectral–spatial feature diffusion (EG-SSFD) model to address this problem. The insight behind this method is that endmember information can be introduced into the diffusion model to enhance feature representation according to the spectral linear mixing model. As shown in Fig. 2(a), the proposed method first employs an autoencoder to extract spectral–spatial features of HSIs. Then, an unmixing module is designed to learn endmember features, providing the spectral prototype representations of HSIs. The diffusion model is subsequently introduced to enhance the spectral–spatial representations by aggregating the endmember information. Unlike existing diffusion-based methods, a denoising network with the cross-attention mechanism is designed to effectively fuse the endmember information. Note that the diffusion process is conducted in the feature space, which improves the computational efficiency of the network. Finally, the updated features are used to reconstruct the original HSIs. Pixelwise anomaly scores are computed based on reconstruction errors.

B. Spectral–Spatial Feature Extraction

The HSI is denoted as $\mathbf{X} \in \mathbb{R}^{H\times W\times C}$, where H, W, and C denote the height, width, and number of spectral bands, respectively. HSIs contain abundant spectral information and rich spatial context information. It is crucial to leverage both spectral characteristics and local spatial context to improve detection accuracy. As shown in Fig. 3, since the anomaly pixel A has similar spectral curve with background pixel B, it is difficult to distinguish A from B. However, the mean spectral curves of their 9×9 neighborhoods are notably different. Therefore, anomaly pixel A can be identified by using the spatial information. To reduce spectral redundancy and improve computational efficiency, we first perform PCA to retain the top C_r principal components that preserve most of the image energy. Next, local spectral–spatial patches are extracted to capture both spectral and spatial information. Using a sliding window, the HSI is represented as a set of spectral–spatial patches centered on each pixel with size $w\times w$, where w denotes the window size, denoted as $\mathbf{X}_p = \{\mathbf{x}_b\}_{b=1}^{N}$, where $N = H \times W$. The set $\mathbf{X}_p$ is then fed into a lightweight convolutional encoder to extract spectral–spatial features. For each local patch, the encoder produces a feature representation $\mathbf{z}_b \in \mathbb{R}^{c\times w\times w}$

$$\mathbf{z}_b = f_{\text{enc}}(\mathbf{x}_b) \tag{1}$$

where c denotes the channel dimension of the intermediate layer of the encoder. Since sampling in diffusion models is time-consuming, the latent features are flattened into vectors to significantly improve the computational efficiency of the diffusion process

$$\widetilde{\mathbf{z}}_b = \text{Flatten}(\mathbf{z}_b) \in \mathbb{R}^{D} \tag{2}$$

where $D = c \times w \times w$. The subsequent diffusion network operates on $\widetilde{\mathbf{z}}_b$ to further model the feature distribution.

C. Endmember Feature Extraction

Hyperspectral linear unmixing aims to estimate the endmembers and their corresponding abundances. Endmembers represent the spectral prototypes of HSIs, while abundances indicate the proportions of different endmembers within a mixed pixel.

Considering the spectral variability in spectral unmixing, a simple yet effective unmixing module is designed to analyze

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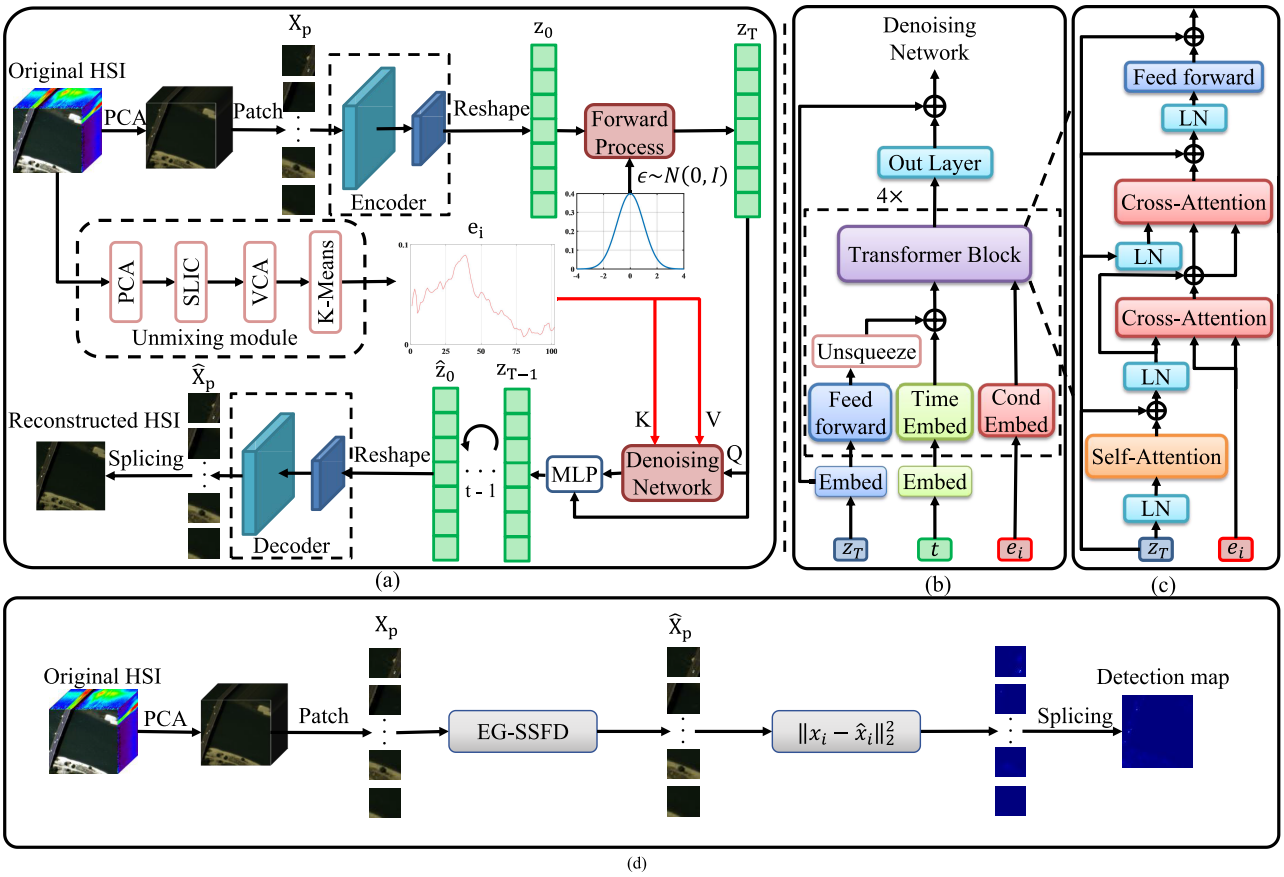
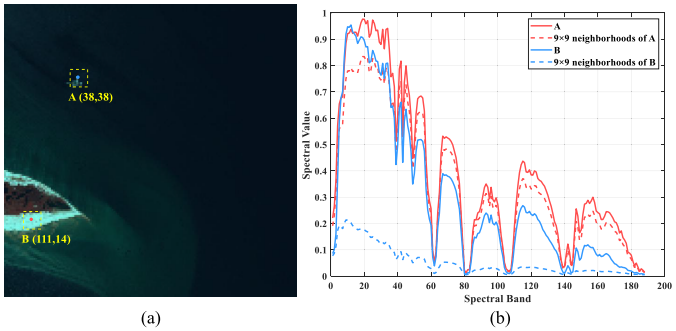


Fig. 2. Flowchart of the proposed method. (a) Proposed network architecture. (b) Denoising network in the diffusion model. (c) Transformer block in the denoising network. (d) Test process.



(a) (b)

Fig. 3. Effectiveness of the spatial information for HAD. (a) False-color image of the Cat Island dataset. (b) Spectral curves of the anomaly pixel A, background pixel B, and their 9×9 neighborhoods.

the components of mixed pixels. Specifically, we use endmember bundles instead of single endmembers to represent the pure materials in HSIs. First, the simple linear iterative clustering (SLIC) [78] method is applied to the HSIs to obtain a set of superpixels. Then, the vertex component analysis (VCA) [81] method is used to extract the endmembers from each superpixel. Next, the endmember bundles are obtained by performing the K-means algorithm on the endmembers. Finally, the mean spectral values of each endmember bundle are used as the initial endmembers, denoted as $\mathbf{E} = \{\mathbf{e}_i \in \mathbb{R}^C\}_{i=1}^q$, where q is the number of endmembers. To adapt the dimensionality of the learned endmember features to the input format required by the attention mechanism, an unsqueeze operation is applied to reshape the endmember features as $\hat{\mathbf{E}} = \{\hat{\mathbf{e}}_i \in \mathbb{R}^{1\times C}\}_{i=1}^q$. These endmember features are then fed into the denoising network to further refine the spectral–spatial features.

D. Feature Diffusion Process

1) Forward Process: Following RX [2], we introduce DDPM in the latent space to better model the complex distribution of normal background. In the forward diffusion process, random Gaussian noise is progressively injected into the latent features to explicitly model the background distribution. Specifically, noise is applied to the flattened latent representation $\tilde{\mathbf{z}}_b$ to transform the initial clean latent distribution into an approximate Gaussian distribution

$$q\left(\tilde{\mathbf{z}}_b^{(t)} \middle| \tilde{\mathbf{z}}_b^{(0)}\right) = \mathcal{N}\left(\tilde{\mathbf{z}}_b^{(t)}; \sqrt{\bar{\alpha}_t}\tilde{\mathbf{z}}_b^{(0)}, (1-\bar{\alpha}_t)\boldsymbol{\chi}\right) \tag{3}$$

where the constant $\bar{\alpha}_t = \prod_{s=1}^t \alpha_s$, α_s is a hyperparameter, and $\boldsymbol{\chi} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ is the standard Gaussian noise. In this way, $\tilde{\mathbf{z}}_b^{(t)}$ can be sampled from the Gaussian distribution by using the reparameterization trick.

2) Denoising Network: In the reverse denoising process, the HSIs features corrupted by Gaussian noise serve as input to the denoising network, which estimates the noise distribution and thus plays a crucial role in the diffusion process.

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Traditional DDPM and its variants often employ U-Net as the backbone of the denoising network. However, significant semantic gaps exist between natural images and HSIs. Moreover, HAD is a typical few-shot task with limited training samples. Therefore, it is necessary to design a denoising network specifically adapted to the HAD task. Additionally, the network should be capable of fusing endmember information to effectively handle mixed pixels. To this end, we design a residual neural network $\epsilon_\theta(\cdot)$ composed of four residual blocks with conditional inputs to estimate the noise distribution, as illustrated in Fig. 2(b).

Before denoising, the latent representations, endmember features, and time embeddings are individually processed by an embedding layer, projecting them to a consistent dimension p. At each time step, the network predicts the noise distribution, and guides the latent representation to gradually revert to its original state. Unlike original diffusion model and its variants that adopt U-Net architectures, our residual-based denoising network is better suited to the latent representations. Furthermore, within each residual block, a cross-attention mechanism is introduced to aggregate the endmember prior extracted from the unmixing module, as shown in Fig. 2(c). During reverse diffusion sampling, the model predicts the noise distribution based on the current noisy state, and the time step is given as follows:

$$\mathbf{z}_{t-1} = \frac{1}{\sqrt{\alpha_t}}\left(\mathbf{z}_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}}\epsilon_\theta(\mathbf{z}_t, t, \hat{\mathbf{E}})\right) + \sigma_t \boldsymbol{\psi}_t \tag{4}$$

where $\epsilon_\theta(\cdot)$ denotes the denoising network, $\beta_t = 1 - \alpha_t$, $\sigma_t = (\beta_t(1-\bar{\alpha}_{t-1})/(1-\bar{\alpha}_t))^{1/2}$, and $\boldsymbol{\psi_t} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ is the standard Gaussian noise.

The loss function for training the denoising network adopts the standard mean squared error form

$$\mathcal{L}_{\text{diff}} = \mathbb{E}_{\mathbf{z}_0,\epsilon,t}\left[\left\|\epsilon - \epsilon_\theta(\mathbf{z}_t, t, \hat{\mathbf{E}})\right\|^2\right]. \tag{5}$$

By incorporating endmember information into the spectral–spatial features, the proposed model effectively improves the reconstruction of complex backgrounds and enhances the response of targets especially those that appear in mixed pixels.

3) Endmember-Guided Feature Diffusion: To incorporate endmember priors and enhance the interaction between feature learning and diffusion, the proposed method designs a Transformer block in each residual denoising network. This block explicitly imposes an endmember-conditional constraint on the spatial-spectral features during the denoising process through the cross-attention mechanism, aiming to reduce impact of mixed pixels on detection performance. The structure of the Transformer block is illustrated in Fig. 2(c), which consists of three hierarchically organized attention layers.

The first layer employs a self-attention mechanism to model internal correlations among local latent features, thereby achieving adaptive representations. Let the input latent feature be denoted as $\mathbf{Z}_t = \{\mathbf{z}_i \in \mathbb{R}^{1\times p}\}_{i=1}^N$. We first apply linear projections to obtain the query, key, and value matrices

$$\mathbf{Q}_1 = \mathbf{W}_{\mathbf{Q}}^{(1)}\mathbf{Z}_t, \quad \mathbf{K}_1 = \mathbf{W}_{\mathbf{K}}^{(1)}\mathbf{Z}_t, \quad \mathbf{V}_1 = \mathbf{W}_{\mathbf{V}}^{(1)}\mathbf{Z}_t \tag{6}$$

where $\mathbf{Q}_1, \mathbf{K}_1, \mathbf{V}_1 \in \mathbb{R}^{N\times 1\times p}$. Then, the attention-enhanced query is computed as follows:

$$\hat{\mathbf{Q}}_1 = \text{Softmax}\left(\frac{\mathbf{Q}_1\mathbf{K}_1^T}{\sqrt{p}}\right)\mathbf{V}_1. \tag{7}$$

This process enhances the spectral consistency and spatial smoothness of local background regions, as the self-attention mechanism captures correlations among local features and adaptively aggregates contextual information.

The second layer utilizes a cross-attention mechanism to integrate the endmember priors. Here, the updated latent features from the first layer serve as the query, while the endmember features $\hat{\mathbf{E}} \in \mathbb{R}^{1\times q\times p}$ are used as key and value

$$\mathbf{Q}_2 = \mathbf{W}_{\mathbf{Q}}^{(2)}\hat{\mathbf{Q}}_1, \quad \mathbf{K}_2 = \mathbf{W}_{\mathbf{K}}^{(2)}\hat{\mathbf{E}}, \quad \mathbf{V}_2 = \mathbf{W}_{\mathbf{V}}^{(2)}\hat{\mathbf{E}} \tag{8}$$

$$\mathbf{A}_2 = \text{Softmax}\left(\frac{\mathbf{Q}_2\mathbf{K}_2^\top}{\sqrt{p}}\right), \quad \hat{\mathbf{Q}}_2 = \mathbf{A}_2\mathbf{V}_2 \tag{9}$$

where $\mathbf{Q}_2 \in \mathbb{R}^{N\times 1\times p}, \mathbf{K}_2, \mathbf{V}_2 \in \mathbb{R}^{1\times q\times p}$. Note that the Softmax function implicitly satisfies both the non-negativity and sum-to-one constraints required by the spectral linear mixing model [80]. Therefore, $\mathbf{A}_2 \in \mathbb{R}^{N\times 1\times q}$ can be interpreted as the abundance estimates for each pixel. Correspondingly, the latent feature $\hat{\mathbf{Q}}_2$ is computed as a linear combination of the endmember features. This step explicitly injects physically interpretable endmember information into the diffusion process, guiding the model to analyze the constituent components of mixed pixels.

The third layer applies a feedback-consistent cross-attention mechanism to correct potential oversmoothing. Here, the original latent feature $\mathbf{Z}_t$ is used as the query, while the fused feature $\hat{\mathbf{Q}}_2$ from the second layer serves as both key and value

$$\mathbf{Q}_3 = \mathbf{W}_{\mathbf{Q}}^{(3)}\mathbf{Z}_t, \quad \mathbf{K}_3 = \mathbf{W}_{\mathbf{K}}^{(3)}\hat{\mathbf{Q}}_2, \quad \mathbf{V}_3 = \mathbf{W}_{\mathbf{V}}^{(3)}\hat{\mathbf{Q}}_2 \tag{10}$$

$$\hat{\mathbf{Q}}_3 = \text{Softmax}\left(\frac{\mathbf{Q}_3\mathbf{K}_3^\top}{\sqrt{p}}\right)\mathbf{V}_3 \tag{11}$$

where $\mathbf{Q}_3, \mathbf{K}_3, \mathbf{V}_3 \in \mathbb{R}^{N\times 1\times p}$. Each attention block is equipped with LayerNorm (LN) and residual connections to maintain stable feature distributions across layers. A feed-forward network (FFN) is applied to the fused latent features to further adjust their consistency.

After the completion of the diffusion process, the updated spectral–spatial features are used to reconstruct the HSI patches $\hat{\mathbf{X}}_p = \{\hat{\mathbf{x}}_b\}_{b=1}^N$ by the decoder. The reconstruction error between the obtained $\hat{\mathbf{X}}_p$ and the original image patches $\mathbf{X}_p$ is calculated to generate an anomaly score map for each image patch. Finally, the anomaly score maps are spliced according to the positions of the image patches, obtaining the complete detection map [81].

E. Joint Training Strategy

The latent diffusion method typically adopts a two-stage training strategy, where the autoencoder or its variants are pretrained and subsequently fine-tuned to the diffusion model. However, since the autoencoder in our model cannot be updated by the diffusion loss, this training paradigm may generate cumulative errors that degrade the detection performance. To address this issue, the proposed method employs a joint

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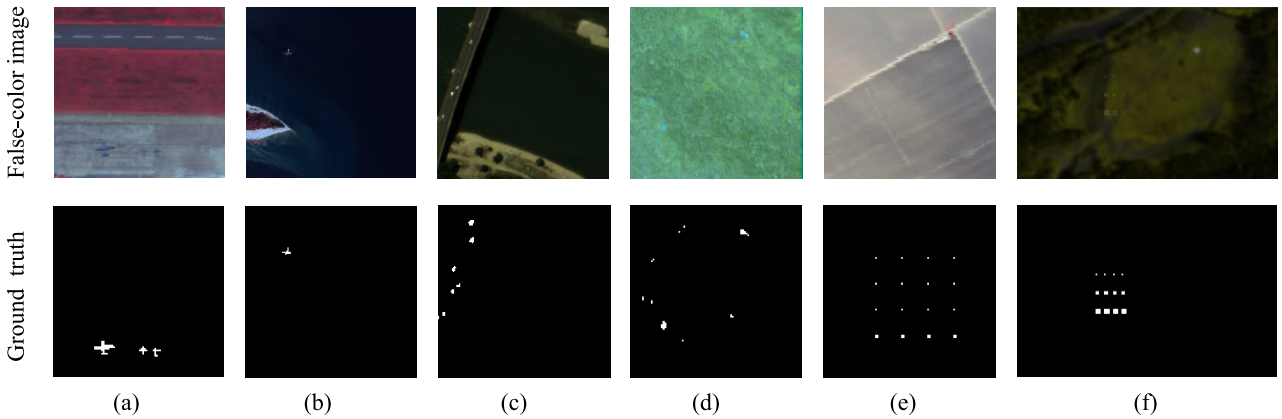

Fig. 4. False-color images and corresponding ground-truth maps of datasets. (a) Gulfport. (b) Cat Island. (c) Pavia. (d) Hyperion. (e) Salinas. (f) HyMap.

TABLE I
DETAILS OF DATASETS

Dataset	Sensors	Spatial Resolution	Wavelength Range	Image Size	Anomaly Size	Anomaly Proportion
Gulfport	AVIRIS	3.4 m/pixel	400–2500 nm	100×100×191	60	0.60%
Cat Island	AVIRIS	17.2 m/pixel	400–2500 nm	150×150×188	19	0.08%
Pavia	ROSIS-03	1.3 m/pixel	430–680 nm	150×150×102	68	0.30%
Hyperion	Hyperion	30.0 m/pixel	400–2500 nm	100×100×145	32	0.32%
Salinas	AVIRIS	1.3 m/pixel	430–860 nm	100×100×204	28	0.28%
HyMap	HyMap	3.0 m/pixel	450–2480 nm	100×150×126	56	0.37%

training strategy that simultaneously trains the autoencoder and the diffusion model. The total loss function of our model consists of the reconstruction loss from the autoencoder and the noise prediction loss from the diffusion model

$$\mathcal{L}_{\text{total}} = \gamma \cdot \mathcal{L}_{\text{rec}} + \sigma \cdot \mathcal{L}_{\text{diff}} \tag{12}$$

where γ and σ are hyperparameters used to balance the contributions of the two loss terms. Their optimal values are tuned via validation in Section IV. Specifically, the reconstruction loss is defined as follows:

$$\mathcal{L}_{\text{rec}} = \frac{1}{N}\sum_{b=1}^{N} \|\mathbf{x_b} - \hat{\mathbf{x}}_\mathbf{b}\|_2^2 . \tag{13}$$

This joint training strategy facilitates mutual enhancement between feature learning and the diffusion process, and reduces the domain shift from natural to hyperspectral imagery. Therefore, it significantly improves the robustness and accuracy of the proposed method.

As shown in Fig. 2(d), each test sample $\mathbf{x_b}$ is input to the proposed model to calculate the reconstruction error $\|\mathbf{x_b} - \hat{\mathbf{x}}_\mathbf{b}\|_2^2$ as its anomaly score at the test stage.

F. Inference Acceleration

Since the feature diffusion of the proposed method is a Markovian process, the running efficiency is much slow. Therefore, we introduce the acceleration strategy in [82] to improve the inference efficiency under the same training process. Specifically, the number of inference steps is evenly divided into three subintervals. Then, uniform sampling is performed in each subinterval to reduce the inference steps. Note that the sampling interval in each subinterval may be different. Since the number of inference steps is much less than that of the forward diffusion steps, the running efficiency will be notably improved.

IV. EXPERIMENTS

A. Experimental Setups

1) Datasets: We use six widely used datasets to evaluate the detection performance of our model. Their false-color images and ground-truth maps are shown in Fig. 4, and detailed information is summarized in Table I. Four real-world datasets include Gulfport, Cat Island, Pavia, and Hyperion. Moreover, two synthetic datasets containing subpixel targets are employed to evaluate the impact of mixed pixels on HAD methods. Several manual subpixel targets are implanted on the cropped Salinas and HyMap datasets according to the spectral linear mixing model [41]. More details of the two synthetic datasets can refer to [17]. Unlike the settings in [17], the distance between adjacent targets is 15 pixels on the Salinas dataset. Furthermore, note that the target abundance is lower in this article on both synthetic datasets. For each column of targets on the HyMap dataset, the abundances are set to 0.05, 0.3, 0.2, and 0.1 from left to right, respectively. For each column of targets on the Salinas dataset, the abundances are set to 0.05, 0.03, 0.04, and 0.01 from left to right, respectively. Therefore, it is more challenging to identify these anomaly targets.

2) Evaluation Metrics: To comprehensively evaluate the performance of the proposed model, we adopt detection map, box plot [83], 2-D receiver operating characteristic (ROC) curve, 3-D ROC curve [84], and the corresponding area under the curve (AUC) metrics. Specifically, both detection map and box plot can measure the background suppression

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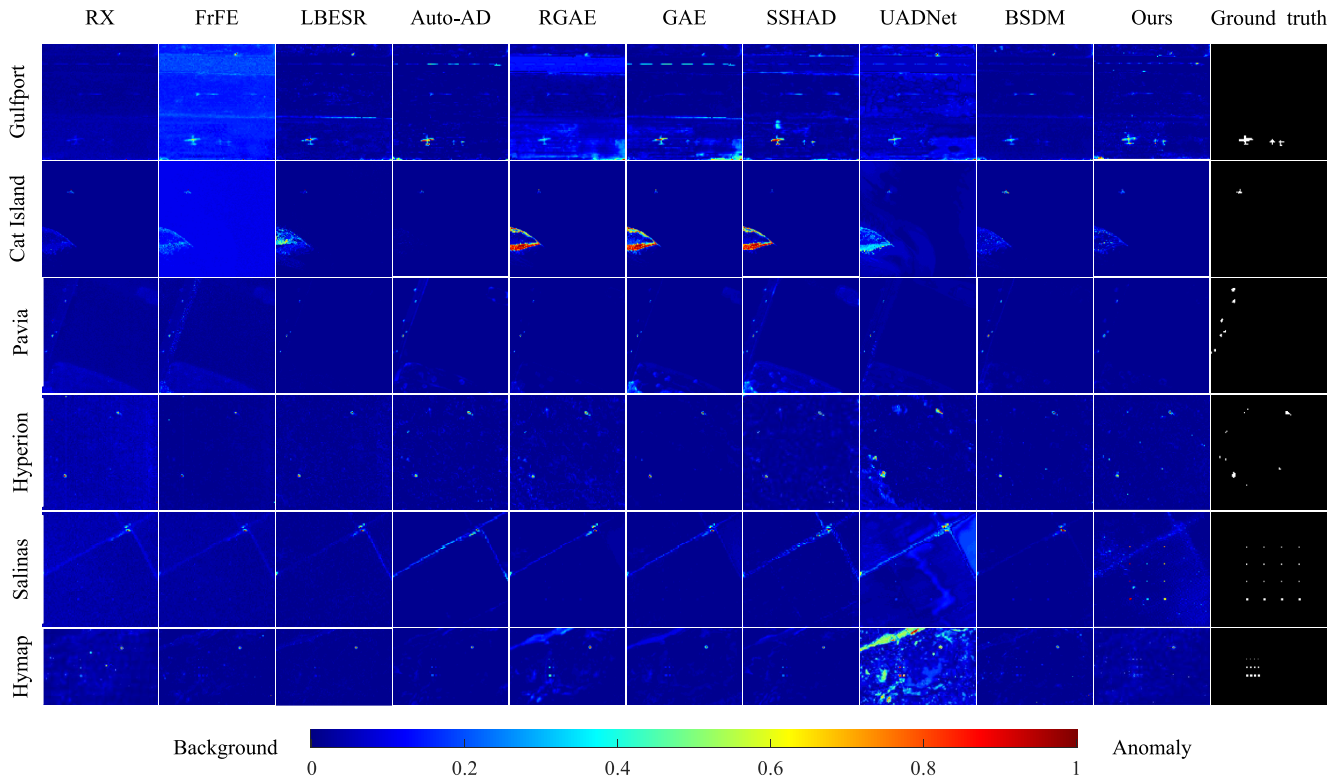

Fig. 5. Detection maps of different methods on all datasets.

performance. Besides, the former can measure the response to the targets. The latter can reflect the separability between anomaly and background. Two-dimensional ROC curve measures the overall balance between the false alarm rate P_f and detection probability P_d [85]. Compared with 2-D ROC curve, 3-D ROC curve is introduced to measure the background suppression performance. AUC(P_d, P_f) values quantitatively measure the overall detection performance. The values closer to 1 indicate the better performance. In addition, AUC(P_d, τ) values reflect the detection response to the targets at varying decision thresholds. The higher values indicate the stronger responses. AUC(P_f, τ) values evaluate the false alarm characteristic, where lower values are desirable as they correspond to fewer false alarms. Here, P_d, P_f, and τ denote the probability of detection, false alarm rate, and detection threshold, respectively.

3) Compared Methods: To validate the effectiveness of the proposed method, we compare it with several state-of-the-art HAD algorithms, including statistical-based methods such as RX and FrFE, representation-based method such as LBESR, and deep learning-based methods such as AutoAD, GAE, RGAE, SSHAD, UADNet, and BSDM. All methods are implemented based on their original articles and official source codes to ensure fair comparison.

The main hyperparameter setting of the proposed method is shown in Table II. During training, the batch size is set to 256, and the model is trained for 100 epochs. The initial learning rate is 1×10^{-3}. It is reduced after 50 epochs by using a learning rate decay schedule. Adam optimizer is used for network training.

TABLE II
HYPERPARAMETER SETTINGS OF OUR METHOD ON DIFFERENT DATASETS

Dataset	C_r	c	w	γ	σ	q
Gulfport	4	4	7	1	0.9	6
Cat Island	4	4	3	1	1	4
Pavia	4	4	3	4	1	4
Hyperion	4	4	7	1	0.6	4
Salinas	4	64	11	5	0.3	4
HyMap	4	4	7	6	0.9	5

B. Experimental Results and Analysis

1) Detection Maps: Detection maps intuitively illustrate the spatial distribution of anomaly responses. As shown in Fig. 5, our method effectively highlights anomaly regions with relatively accurate localization and clear boundaries. Especially on the two synthetic datasets with subpixel targets, our method obtains more intense target response than other methods. Meanwhile, our method achieves a good balance between anomaly enhancement and background suppression. Although some residual background responses remain in complex areas, the overall background noise is effectively reduced. In contrast, traditional methods such as RX and FrFE exhibit higher background noise and blurred anomaly boundaries. Some deep learning models such as SSHAD and UADNet suffer from numerous false alarms due to insufficient spatial modeling capability.

2) Box Plots: To quantify the separability between background and anomalies, box plots of different methods on the six datasets are shown in Fig. 6. The boxes represent the 10%

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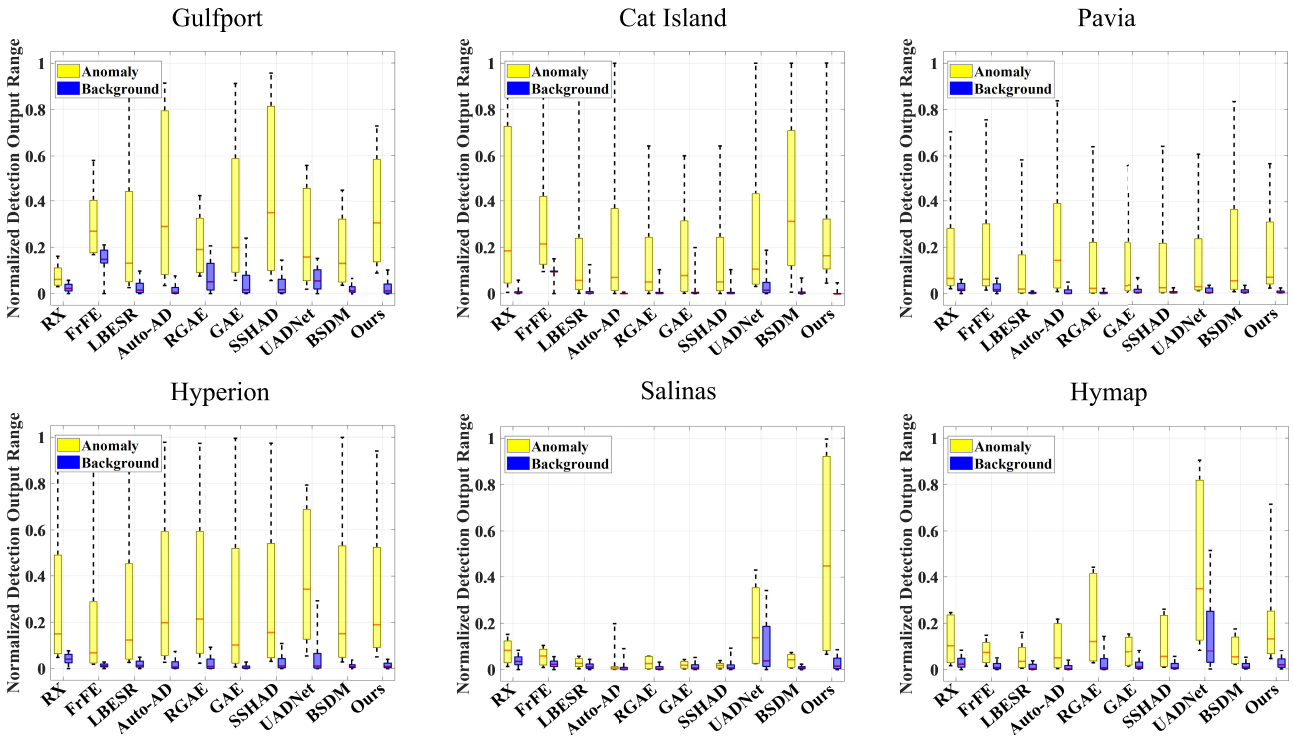

Fig. 6. Box plots of different methods.

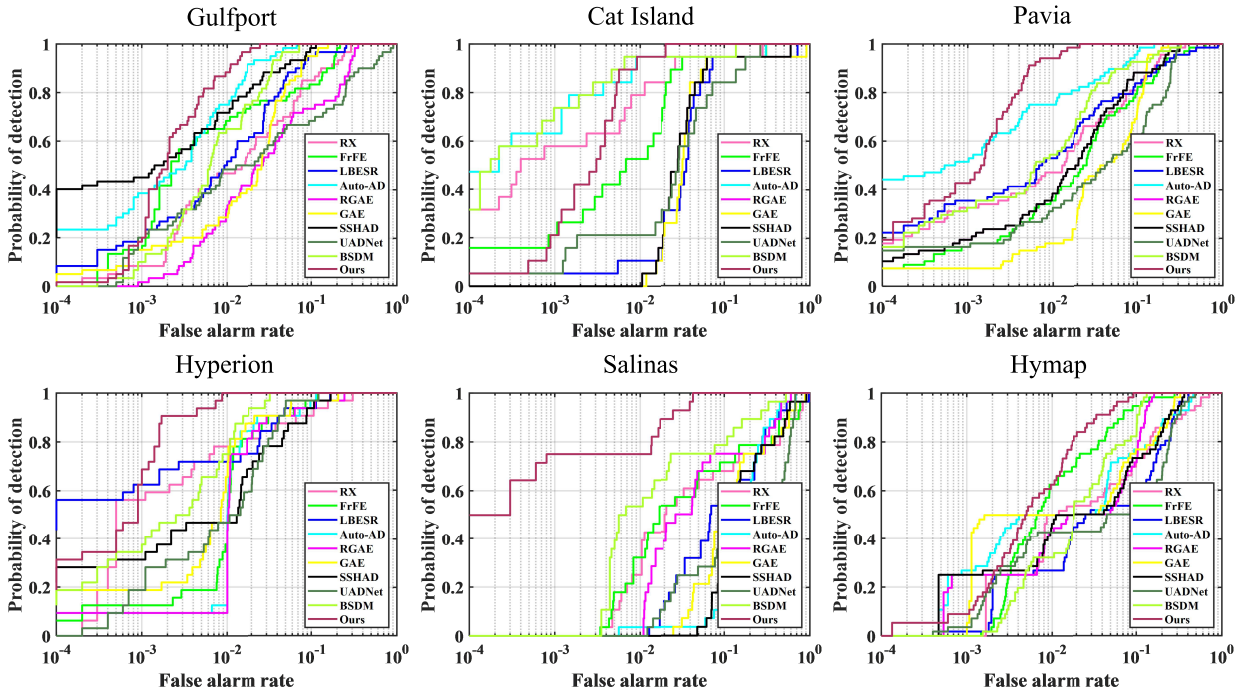

Fig. 7. 2-D ROC curves of (P_d, P_f) for different methods.

to 90% percentile range, and the whiskers indicate the full data range. Our method yields highly compact background distributions and distinctly higher anomaly scores. On the Cat Island, Pavia, and Hyperion datasets, the maximum background score is lower than the minimum anomaly score, demonstrating excellent anomaly–background separability. Note that, due to the extremely low abundances of targets, most methods cannot separate background from anomaly on the Salinas and HyMap datasets. However, our method also has higher degree of separability on the two datasets. This property improves threshold robustness and enhances localization precision of targets.

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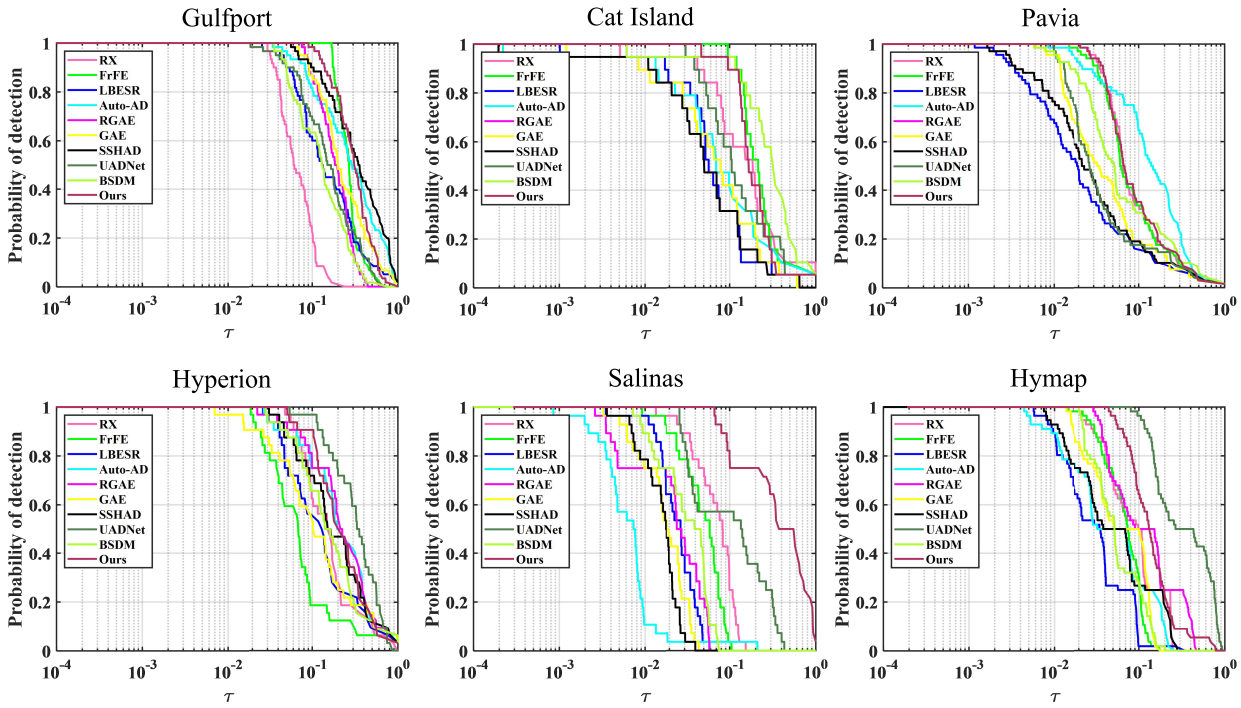

Fig. 8. 2-D ROC curves of (P_d, τ) for different methods.

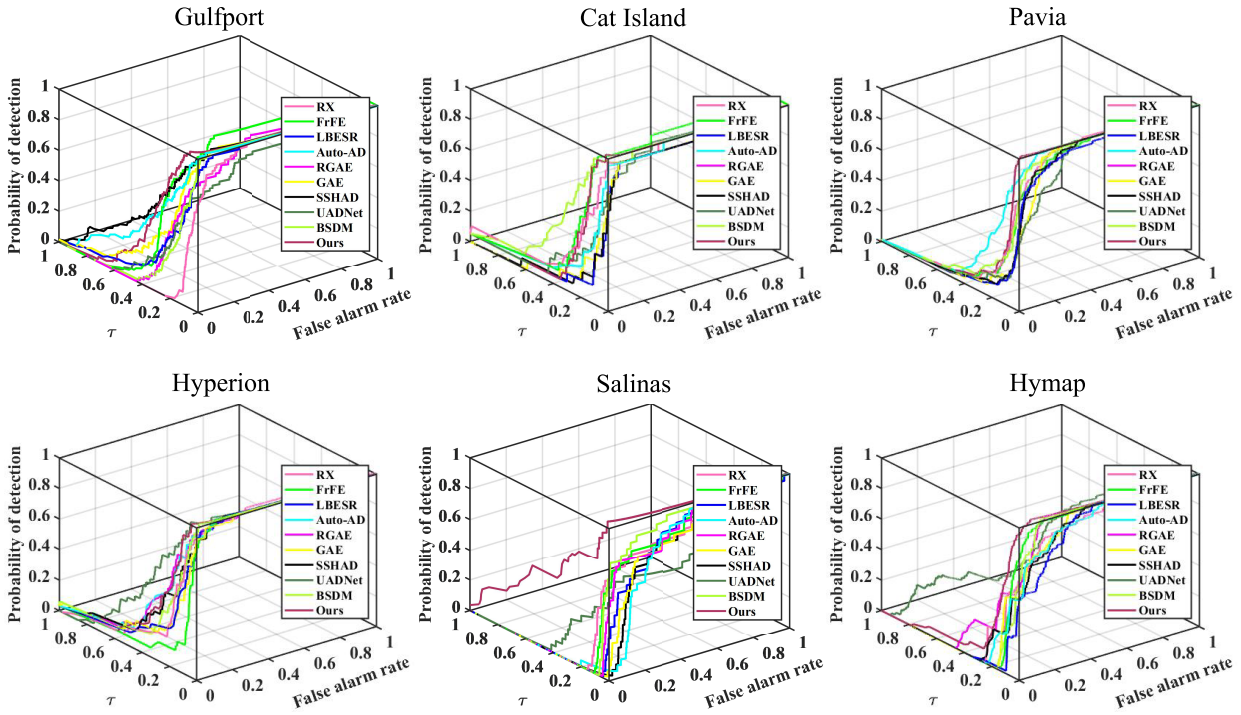

Fig. 9. 3-D ROC curves of different methods.

3) ROC Curves: Two-dimensional ROC curves of (P_d, P_f) are shown in Fig. 7. The curves achieved by our method are the closest to the top-left corner on the Gulfport, Pavia, Hyperion, Salinas, and HyMap datasets, indicating the best overall performance. On the Cat Island dataset, our method performs comparable to BSDM. Moreover, as shown in Fig. 8, the 2-D ROC curves of (P_d, τ) of our method are closer to the top-right corner on the Gulfport, Cat Island, Pavia, and Salinas datasets, which reflects that our method achieves higher detection probabilities at varying decision thresholds. Although BSDM slightly outperforms our method in the initial segment of the 2-D ROC curves of (P_d, P_f) on the Cat Island dataset, our method shows better performance in terms of 2-D ROC curves of (P_d, τ). Therefore, our method will achieve higher AUC(P_d,

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TABLE III
AUC VALUES OF DIFFERENT METHODS ON ALL DATASETS

Datasets	AUC Values	RX	FrFE	LBESR	AutoAD	RGAE	GAE	SSHAD	UADNet	BSDM	Ours
Gulfport	AUC(P_d, P_f)	0.9526	0.9641	0.9719	0.9925	0.9182	0.9685	0.9870	0.9664	0.9865	**0.9962**
	AUC(P_d, τ)	0.0736	0.2794	0.2192	0.3127	0.2034	0.2915	**0.4210**	0.2023	0.1617	0.3387
	AUC(P_f, τ)	0.0248	0.1534	0.0221	0.0326	0.0664	0.0347	0.0279	0.0584	**0.0184**	0.0211
Cat Island	AUC(P_d, P_f)	0.9808	0.9741	0.9304	0.9826	0.9406	0.9219	0.9406	0.9467	0.9918	**0.9961**
	AUC(P_d, τ)	0.2496	0.2729	0.1273	0.1786	0.1058	0.1267	0.1058	0.1985	**0.3672**	0.2301
	AUC(P_f, τ)	0.0065	0.0976	0.0099	0.0206	0.0142	0.0187	0.0142	0.0272	0.0071	**0.0035**
Pavia	AUC(P_d, P_f)	0.9538	0.9354	0.9409	0.9855	0.9529	0.9386	0.9529	0.9026	0.9720	**0.9975**
	AUC(P_d, τ)	0.1343	0.1335	0.0756	**0.2014**	0.0860	0.0926	0.0860	0.0909	0.1369	0.1362
	AUC(P_f, τ)	0.0233	0.0228	**0.0027**	0.0056	0.0028	0.0080	0.0190	0.0073	0.0077	0.0033
Hyperion	AUC(P_d, P_f)	0.9768	0.9827	0.9844	0.9818	0.9767	0.9836	0.9755	0.9812	0.9938	**0.9988**
	AUC(P_d, τ)	0.2112	0.1295	0.2065	0.2916	0.2968	0.2118	0.2649	**0.3814**	0.2398	0.2739
	AUC(P_f, τ)	0.0424	0.0144	0.0169	0.0175	0.0216	**0.0069**	0.0206	0.0295	0.0114	0.0129
Salinas	AUC(P_d, P_f)	0.8249	0.8431	0.8272	0.9044	0.8784	0.7966	0.7833	0.8992	0.9366	**0.9940**
	AUC(P_d, τ)	0.0782	0.0552	0.0273	0.0140	0.0291	0.0205	0.0176	0.1578	0.0375	**0.4855**
	AUC(P_f, τ)	0.0382	0.0250	0.0150	0.0078	**0.0069**	0.0115	0.0137	0.0704	0.0093	0.0228
HyMap	AUC(P_d, P_f)	0.9028	0.9751	0.8954	0.9224	0.9480	0.9299	0.9205	0.8924	0.9634	**0.9868**
	AUC(P_d, τ)	0.1112	0.0753	0.0450	0.0775	0.1674	0.0790	0.0867	**0.4340**	0.0676	0.1718
	AUC(P_f, τ)	0.0284	0.0134	0.0088	**0.0078**	0.0199	0.0145	0.0101	0.1188	0.0123	0.0234

τ) values, demonstrating intense target responses. Furthermore, the 3-D ROC curves in Fig. 9 show that our method achieves satisfactory detection probability on all datasets. Meanwhile, it can better suppress the background on the Hyperion and HyMap datasets.

4) AUC Values: The AUC values for all detection algorithms on different datasets are reported in Table III. The best AUC values are given in bold, and the second-best results are underlined. Based on the table, we draw the following observations.

1) Overall, our method significantly outperforms all compared methods in terms of AUC(P_d, P_f) values on the six datasets. Specifically, compared with the second-best results, our method improves the AUC(P_d, P_f) values by 0.37%, 0.43%, 1.20%, 0.50%, 5.74%, and 1.17%, respectively. This demonstrates the superior robustness of our method to both full-pixel and subpixel anomaly detection.
2) LBESR adopts a clustering-based strategy to extract background endmembers. However, its AUC(P_d, P_f) values are much lower than ours on the Salinas and HyMap datasets with severe spectral mixing. LBESR conducts detection in the raw HSI space, where the high similarity between anomaly and background spectral curves tends to contaminate the endmember estimation, leading to degradation of detection accuracy.
3) UADNet introduces spectral unmixing mechanism to enhance the representation of the model. Nevertheless, it uses abundance maps as the input of the deep clustering module to estimate density. When the target abundances are extremely low on the datasets, its clustering performance will degrade. Therefore, UADNet achieves inferior detection performance to our method on the Salinas and HyMap datasets in terms of AUC(P_d, P_f) values. That is because, our method introduces endmember-conditional attention into the diffusion process according to the linear mixing model, which enhances the feature representation.
4) Although BSDM also introduces diffusion process for HAD, its AUC(P_d, P_f) values are much lower than ours on all datasets. BSDM performs diffusion in the raw image space while our methods perform diffusion in the feature space. Moreover, BSDM only uses spectral features of HSIs, while our method fuses the spatial–spectral features with endmember features to improve the feature representations.
5) As shown in Table III, our method achieves the highest AUC(P_d, τ) values on the Gulfport, Cat Island, Pavia, and Salinas datasets, which shows superior detection accuracy on these datasets. Although the AUC(P_d, τ) values are slightly lower on other datasets, our method exhibits better overall detection performance in terms of AUC(P_d, P_f) values. In particular, our method significantly outperforms other methods on the Salinas dataset with much lower target abundances, demonstrating stronger responses to subpixel targets.
6) The AUC(P_f, τ) values demonstrate that our method achieves competitive performance of background suppression on most datasets. In particular, our method achieves the lowest AUC(P_f, τ) value on the synthetic HyMap dataset with subpixel targets, indicating a stronger capability of suppressing false alarms.

C. Model Analyses

1) Analysis of the Training Paradigms: In addition, to validate the effectiveness of the proposed joint training strategy, we conduct experiments on all datasets to compare the two-stage training approach with joint training approach. As shown in Table IV, our method achieves higher AUC values on all datasets when using joint training. These results

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TABLE IV
COMPARISON OF AUC(P_d, P_f) VALUES BETWEEN JOINT TRAINING AND TWO-STAGE TRAINING ON ALL DATASETS

Training Paradigm	Gulfport	Cat Island	Pavia	Hyperion	Salinas	HyMap
Two-stage Training	0.9831	0.9928	0.9911	0.9828	0.9632	0.9499
Joint Training	**0.9962**	**0.9961**	**0.9975**	**0.9988**	**0.9940**	**0.9868**

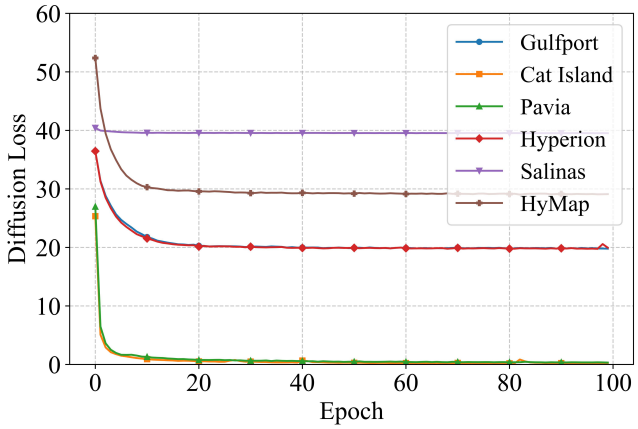

Fig. 10. Diffusion loss curve on all datasets.

TABLE V
IMPACT OF DIFFERENT ABUNDANCE LEVELS ON OUR METHOD

Dataset	0.001	0.01	0.05	0.1	0.3
Salinas	0.9426	0.9939	0.9976	0.9956	0.9993
HyMap	0.9409	0.9492	0.9868	0.9887	0.9885

indicate that the end-to-end joint optimization strategy effectively enhances the representation capability of the model. That is because, joint optimization reduces the cumulative errors between the feature learning and diffusion denoising process. Thereby, it enables more robust and accurate anomaly detection.

Moreover, we visualize the diffusion training loss of the proposed network to demonstrate its convergence. The number of iterations of the proposed network is set to 100 on all datasets. As shown in Fig. 10, the diffusion loss values decrease first and then tend to stabilize on all datasets. Therefore, the diffusion training converges well.

2) Analysis of the Target Abundance: We replace the lowest target abundances on the two synthetic datasets with different abundances to investigate their effect on the detection performance. As shown in Table V, the AUC(P_d, P_f) values tend to increase as the target abundances increase on these datasets. That is because, the larger target abundances reflect the larger spectral differences between background and anomalies.

3) Analysis of the Number of Cross-Attention Layers: As shown in Table VI, the number of cross-attention layers in the proposed noise estimation network has remarkable effect on the AUC(P_d, P_f) values. Too small number of cross-attention layers (e.g., 1) cannot achieve optimal AUC(P_d, P_f) values due to the limited representation capability. Moreover, too large number of cross-attention layers will result in long running time of the model, which limits its applications in real-time scenarios.

TABLE VI
IMPACT OF DIFFERENT NUMBER OF CROSS-ATTENTION LAYERS ON OUR METHOD

Dataset	1	2	3	4
Gulfport	0.9958	**0.9962**	0.9944	0.9939
Cat Island	0.9935	**0.9961**	0.9933	0.9934
Pavia	0.9966	**0.9975**	0.9964	0.9964
Hyperion	0.9974	**0.9988**	0.9984	0.9947
Salinas	0.9912	**0.9940**	0.9915	0.9928
HyMap	0.9744	**0.9868**	0.9798	0.9763

4) Analysis of the Robustness to Noise: To investigate the effect of noise of HSIs on the detection performance, we add the Gaussian noise to the Hyperion and Salinas datasets, and apply the proposed model to these datasets with different levels of signal-to-noise ratio (SNR). As shown in Table VII, the AUC(P_d, P_f) values tend to increase as the SNR increases. Therefore, the Gaussian noise plays a negative effect on the detection performance of our method.

5) Analysis of the Quality of Initial Endmembers: The quality of initial endmembers can affect the detection performance of our method. Since the VCA method uses random numbers to calculate the endmembers, the endmembers may be different when performing several times of VCA. We compare the detection performance of running VCA once and iteratively running VCA in the training process without acceleration. As shown in Table VIII, the AUC(P_d, P_f) values are slightly higher when iteratively running VCA, because it obtains more stable initial endmembers. However, the running time will be significantly reduced by performing VCA once before the network training. Considering the running efficiency, we obtain the endmembers before the network training.

D. Ablation Study

To evaluate the contribution of unmixing module, diffusion module, the spatial feature learning, and acceleration inference, we conduct ablation study on several variants. Specifically, we set a fully connected autoencoder (FCAE) and a ConvAE as baselines. Both of them use reconstruction errors as anomaly scores. Then, the diffusion process, endmember extraction (including VCA-based method and endmember-bundle (EB)-based method), and inference acceleration are progressively introduced to evaluate the performance improvements of each module. The AUC(P_d, P_f) values on all datasets are reported in Table IX.

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TABLE VII
IMPACT OF DIFFERENT SNR LEVELS ON OUR METHOD

Dataset	10 dB	15 dB	20 dB	30 dB	40 dB	50 dB	60 dB
Hyperion	0.8050	0.9374	0.9876	0.9983	0.9982	0.9985	0.9985
Salinas	0.5086	0.8147	0.8987	0.9838	0.9930	0.9932	0.9937

TABLE VIII
IMPACT OF DIFFERENT QUALITY OF THE INITIAL ENDMEMBER ON OUR METHOD

Initial endmembers	Gulfport		Cat Island		Pavia		Hyperion		Salinas		HyMap	
	AUC	Time	AUC	Time	AUC	Time	AUC	Time	AUC	Time	AUC	Time
Running VCA iteratively	0.9969	246.19	0.9961	506.13	0.9981	859.11	0.9980	154.15	0.9876	305.42	0.9883	491.16
Running VCA once	0.9953	116.24	0.9953	394.13	0.9965	346.38	0.9968	111.58	0.9837	247.76	0.9857	279.73

TABLE IX
AUC(P_d, P_f) VALUES ACHIEVED BY DIFFERENT VARIANTS OF OUR METHOD ON ALL DATASETS

Variants	Gulfport	Cat Island	Pavia	Hyperion	Salinas	HyMap
FCAE	0.9425	0.9657	0.9596	0.9760	0.8374	0.8890
FCAE + Diffusion	0.9613	0.9685	0.9630	0.9769	0.8791	0.9129
FCAE + Diffusion + VCA	0.9657	0.9715	0.9656	0.9919	0.8915	0.9140
FCAE + Diffusion + EB	0.9823	0.9752	0.9695	0.9936	0.9312	0.9662
FCAE + Diffusion + VCA + Acceleration	0.9682	0.9689	0.9652	0.9937	0.9101	0.9269
FCAE + Diffusion + EB + Acceleration	0.9847	0.9703	0.9738	0.9950	0.9318	0.9667
ConvAE	0.9899	0.9803	0.9907	0.9928	0.9428	0.9252
ConvAE + Diffusion	0.9910	0.9915	0.9916	0.9937	0.9566	0.9738
ConvAE + Diffusion + VCA	0.9953	0.9953	0.9965	0.9968	0.9837	0.9857
ConvAE + Diffuion + EB	0.9955	0.9953	0.9970	0.9979	0.9921	0.9843
ConvAE + Diffuion + VCA + Acceleration	0.9959	0.9954	0.9965	0.9978	0.9846	0.9858
Ours	**0.9962**	**0.9961**	**0.9975**	**0.9988**	**0.9940**	**0.9868**

1) Effect of the Diffusion Module: To evaluate the effectiveness of diffusion module, we remove the cross-attention mechanism from the proposed denoising network that is used to fuse the endmember features. Compared with FCAE, adding the diffusion module improves the AUC(P_d, P_f) values by 1.88%, 0.28%, 0.34%, 0.09%, 4.17%, and 2.39% on the six datasets, respectively. Compared with ConvAE, the corresponding improvements of AUC(P_d, P_f) values are 0.11%, 1.12%, 0.09%, 0.09%, 1.38%, and 4.86% on the six datasets, respectively. The improvement is more prominent especially on the two synthetic datasets due to the excellent generative capability of diffusion module. This validates that the proposed diffusion module is well-suitable for HSIs, and enhances the feature representation of the models.

2) Effect of the Unmixing Module: The AUC(P_d, P_f) values are further improved on all datasets by introducing endmember-conditioned attention mechanism into the diffusion module, which demonstrates that explicitly incorporating endmember priors guides the model to distinguish anomalies from complex backgrounds. The endmember-conditioned attention mechanism enforces physical constraints to preserve the spectral structures of latent representations, and facilitates interactions between the learned features and endmember features. Therefore, it can improve the contrast between anomaly and background. Furthermore, it can be observed that the EB-based method achieves higher AUC(P_d, P_f) values than the VCA-based method when exploiting FCAE to learn the spectral features. The AUC values are improved by 1.66%, 0.37%, 0.39%, 0.17%, 3.97%, and 5.22% on the six datasets, respectively. If the spatial features are extracted by ConvAE, the EB-based method achieves comparable performance with VCA-based method, with slight improvements of 0.03%, 0.05%, 0.11%, and 0.84% on the Gulfport, Pavia, Hyperion, and Salinas datasets, respectively. That is because, the spatial contextual information can help reduce the negative effect of spectral variability.

3) Effect of the Spatial Feature: Due to the spatial continuity of background, spatial information is useful to distinguish background from anomaly. As shown in Table IX, the AUC(P_d, P_f) values achieved by the FCAE-based variants are lower than those achieved by the ConvAE-based variants on all datasets. Note that, since the target abundances are extremely low on the Salinas and HyMap datasets, the targets cannot be detected only by spectral features. Therefore, the FCAE-based variants achieve poor detection performance on these datasets. However, our method can better recognize the subpixel targets under the ConvAE framework. This indicates that the convolution structure effectively captures local spatial–spectral correlations and improves the discrimination ability of features.

4) Effect of the Inference Acceleration: We investigate the effect of the inference acceleration strategy on the proposed method by setting different numbers of uniform sampling steps

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TABLE X
IMPACT OF DIFFERENT INFERENCE STEPS ON ALL DATASETS

Inference steps	Gulfport		Cat Island		Pavia		Hyperion		Salinas		HyMap	
	AUC	Time	AUC	Time	AUC	Time	AUC	Time	AUC	Time	AUC	Time
(10,30,10)	0.9961	96.62	0.9961	186.44	0.9975	149.52	0.9985	87.89	0.9937	215.80	0.9814	120.15
(10,30,30)	0.9962	98.96	0.9960	190.85	0.9975	181.46	0.9986	98.76	0.9936	217.93	0.9816	183.27
(10,30,60)	0.9962	102.60	0.9960	202.28	0.9975	194.23	0.9987	113.33	0.9935	225.51	0.9848	198.28
(20,30,10)	0.9960	97.36	0.9960	197.28	0.9975	188.88	0.9986	88.54	0.9940	217.70	0.9818	121.57
(20,30,30)	0.9961	100.57	0.9960	217.54	0.9975	195.99	0.9984	92.571	0.9939	220.09	0.9823	169.85
(20,30,60)	0.9961	104.12	0.9960	225.01	0.9975	210.82	0.9987	108.09	0.9940	224.18	0.9828	229.91
(30,30,10)	0.9962	98.98	0.9960	192.81	0.9975	181.75	0.9986	85.42	0.9939	222.37	0.9828	223.97
(30,30,30)	0.9961	101.27	0.9960	199.65	0.9975	190.86	0.9985	91.66	0.9938	217.96	0.9838	252.21
(30,30,60)	0.9962	104.89	0.9960	206.68	0.9975	221.59	0.9988	108.26	0.9936	229.17	0.9868	278.25

TABLE XI
RUNNING TIME (SECOND) OF DIFFERENT METHODS

Method	Gulfport	Cat Island	Pavia	Hyperion	Salinas	HyMap
RX	0.22	0.96	0.68	0.17	0.17	0.17
FrFE	7.71	16.02	7.46	5.36	8.03	6.73
LBESR	0.64	1.97	1.57	0.61	0.67	0.86
Auto-AD	18.07	21.90	29.15	27.53	28.41	29.09
RGAE	50.84	89.69	64.37	44.05	52.14	52.20
GAE	22.41	79.76	35.23	24.78	22.43	25.83
SSHAD	17.13	17.98	19.67	18.44	18.39	17.38
UADNet	11.09	16.11	13.19	9.88	10.71	13.24
BSDM	8.47	12.75	12.32	8.52	9.01	10.39
Ours w/o Acceleration	204.29	423.16	407.85	144.88	282.02	330.96
Ours	98.96	186.44	149.52	108.26	217.70	278.25

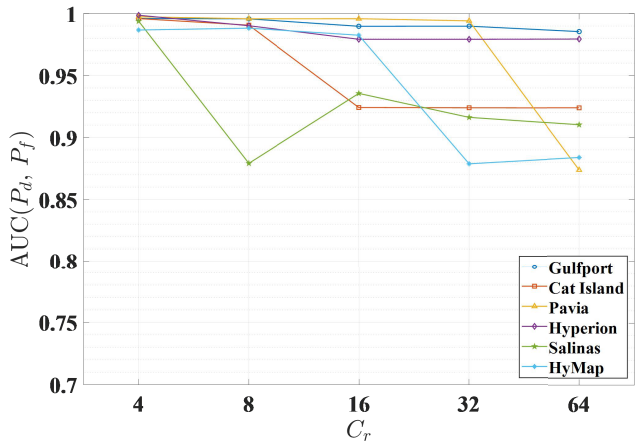

Fig. 11. Impact of parameter C_r.

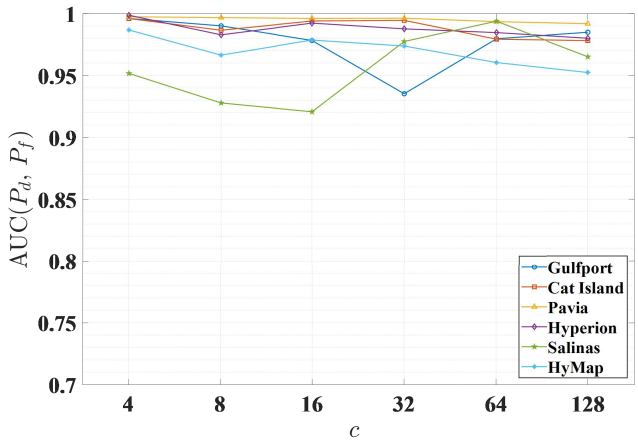

Fig. 12. Impact of parameter c.

in the three subintervals of total inference steps. It can be observed from Table IX that the acceleration strategy does not notably sacrifice the detection accuracy. There are even slight improvements of AUC(P_d, P_f) values on the Hyperion and Salinas datasets. Moreover, note that (10, 30, 10) in Table X denotes that the number of uniform sampling in the three subintervals are 10, 30, and 10, respectively. As shown in Table X, the impact of difference inference steps on detection accuracy is relatively small on all datasets. However, the inference steps impose a significant impact on the running efficiency of the proposed method.

E. Parameter Sensitivity Analyses

1) Impact of C_r: The impact of hyperparameter C_r on the detection performance is shown in Fig. 11. We set the range of C_r to $\{4, 8, 16, 32, 64\}$. The hyperparameter C_r is used to adjust the number of principal components of PCA. Therefore, it controls the spectral dimension of the input features. Too large values of C_r will result in band redundancy and performance degradation. Lower value of C_r is preferable for most datasets.

2) Impact of c: The impact of hyperparameter c on the detection performance is shown in Fig. 12. Hyperparameter c controls the dimension of the output features of the

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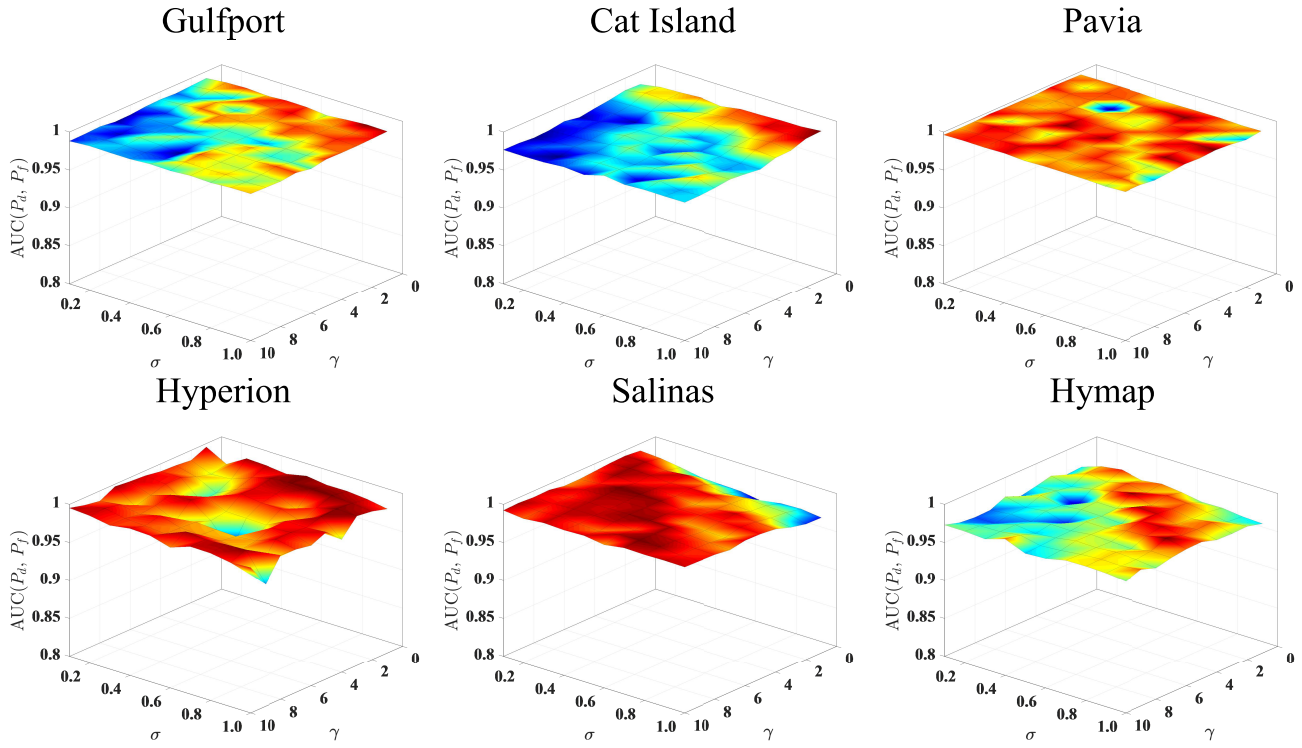

Fig. 13. Impact of parameters γ and σ.

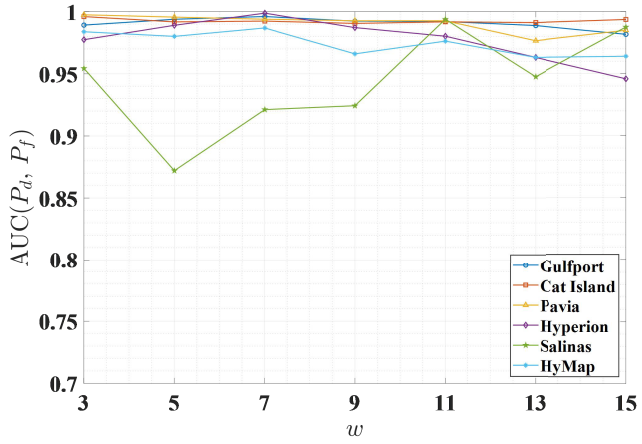

Fig. 14. Impact of parameter w.

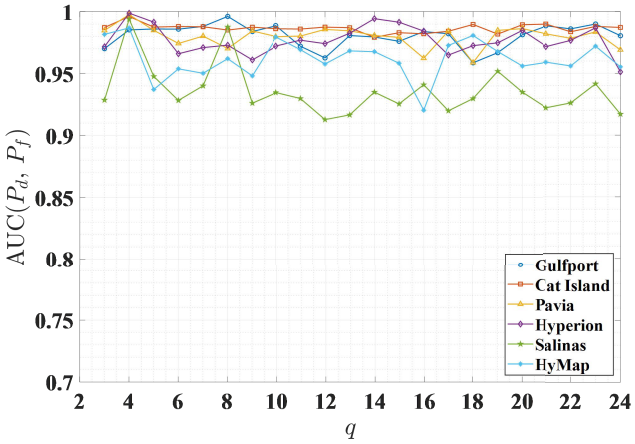

Fig. 15. Impact of parameter q.

encoder, thereby affecting the representation capability of the subsequent diffusion module. We set the range of c to $\{4, 8, 16, 32, 64, 128\}$. When the value of c is large, the network training is time-consuming. Lower value of c is preferable for most datasets.

3) Impact of γ and σ: The impact of hyperparameters γ and σ on detection performance is shown in Fig. 13. The ranges of γ and σ are set to $\{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$ and $\{0.1, 0.2, 0.3, 0.4\ , 0.5, 0.6, 0.7, 0.8, 0.9, 1.0\}$, respectively. It can be seen that the AUC(P_d, P_f) values fluctuate significantly on the Gulfport and Salinas datasets. The optimal combinations of (γ, σ) are obtained according to differences in HSI background structures.

4) Impact of w: The impact of hyperparameter w on detection performance is illustrated in Fig. 14. The range of w is set to $\{3, 5, 7, 9, 11, 13, 15\}$. This hyperparameter plays a crucial role in spatial feature extraction. The results show that w significantly affects the AUC(P_d, P_f) values particularly on the Salinas dataset. That is because, our method requires sufficient spatial information to identify the targets with extremely low abundances. The value of 7 is preferable for most datasets.

5) Impact of q: The impact of parameter q on detection performance is shown in Fig. 15. The range of q is set from 2 to 24 with an interval of 2. Hyperparameter q represents the number of endmembers that are used in the unmixing module. This parameter plays a crucial role in learning accurate spectral characteristics of HSIs, thereby influencing the accuracy and

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robustness of anomaly detection. This value being lower than 10 is preferable according to the sparsity of endmembers.

F. Running Time

The running time of all HAD algorithms is presented in Table XI. All methods are performed on a desktop equipped with 32 GB of memory and an NVIDIA RTX 3080 GPU. RX, FrFE, LBESR, RGAE, and GAE follow their official implementations using MATLAB R2018b. The other deep learning-based methods are implemented by using PyTorch. Since RX calculates anomaly scores of HSIs in a single computation, it runs the fastest on all datasets. Compared with other methods, our method spends more running time due to the sampling process of the diffusion model and the cross-attention mechanism in the denoising process. Although BSDM introduces the diffusion process, it is based on the FC network and does not use the attention mechanism. Therefore, BSDM takes less running time than our method. Note that, the inference acceleration strategy notably improves the running time of our method. However, our algorithm achieves superior detection accuracy.

V. Conclusion and Discussion

In this article, we proposed an endmember-guided feature diffusion model for HAD. Our method designs a diffusion model to regularize the HSI feature distribution to improve the representation capability of the network. Moreover, an unmixing module is introduced to extract the background endmember to update the HSI features so that our method can better identify subpixel targets. Experimental results show that our method achieves superior detection performance to the existing state-of-the-art methods especially on the datasets with subpixel targets that have extremely low abundances.

Nevertheless, our method is time-consuming due to the quadratic complexity of the Transformer and the inherent inference process of the diffusion model. Moreover, our method may suffer from performance degradation in the noisy scene or the spectral nonlinear mixing scene caused by intimate mixture and scattering effect. In the future, we will study a more efficient attention mechanism in Transformer to reduce the training time of our method. Furthermore, we will try to introduce a nonlinear unmixing method to obtain the endmembers, and combine the denoising approach to improve the detection performance in highly noisy scene. However, our method provides a fundamental baseline that introduces the diffusion model to detect subpixel targets.

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Caihuan Huang received the B.E. degree in communication engineering from Guizhou University, Guiyang, China, in 2024, where he is currently pursuing the M.E. degree in communication engineering.

His research interests include hyperspectral image processing and deep learning.

Li Liu received the Ph.D. degree in information and communication engineering from the National University of Defense Technology (NUDT), Changsha, China, in 2012.

During her Ph.D. study, she spent more than two years as a Visiting Student with the University of Waterloo, Waterloo, ON, Canada, from 2008 to 2010. From 2015 to 2016, she spent ten months visiting the Multimedia Laboratory with The Chinese University of Hong Kong, Hong Kong. From December 2016 to November 2018, she worked as a Senior Researcher with the Machine Vision Group, University of Oulu, Oulu, Finland. She is currently a Full Professor with NUDT. Her papers have currently over more than 7700 citations in Google Scholar. Her research interests include computer vision, machine learning, artificial intelligence, trustworthy AI, and synthetic aperture radar.

Chao Xiao received the B.E., M.E., and Ph.D. degrees from the National University of Defense Technology (NUDT), Changsha, China, in 2016, 2018, and 2023, respectively.

He is currently a Lecturer with the College of Electronic Science and Technology, NUDT. His recent research interests focus on small target detection and tracking.

Qiang Ling received the B.Eng. degree in measurement engineering and the M.Eng. degree in control science and engineering from Air Force Engineering University, Xi'an, Shanxi, China, in 2013 and 2016, respectively, and the Ph.D. degree in information and communication engineering from the National University of Defense Technology (NUDT), Changsha, China, in 2019.

He is currently a Lecturer with the College of Electronic Science and Technology, NUDT. His research interests include pattern recognition and hyperspectral image processing.

Guosheng Li received the B.E. degree in data science and big data technology from Guizhou University, Guiyang, China, in 2024, where he is currently pursuing the M.E. degree in electronic information.

His research interests include hyperspectral image processing and deep learning.

Kun Li received the B.S. degree in computer science and technology from China University of Mining and Technology, Xuzhou, China, in 2007, the M.S. degree in computer application technology from Tianjin University of Technology, Tianjin, China, in 2010, and the Ph.D. degree in information and communication engineering from the National University of Defense Technology, Changsha, China, in 2023.

He is currently a Lecturer with the College of Big Data and Information Engineering, Guizhou University, Guiyang, China. His research interests include pattern recognition and hyperspectral image processing.

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